PROJECT TECHNICAL SPECIFICATION & ARCHITECTURE WHITEPAPER



MilkDirect

Farm-to-Door Fresh Milk Marketplace

A full-stack, hyperlocal direct-to-consumer dairy ecosystem connecting cattle-owning farmers and village milkmen directly with residential consumers. Features strict role-based access control, automated recurring subscription scheduling, direct zero-commission UPI payouts, and complete quality metric transparency.

Flutter 3.3+ (Mobile & Web) | Supabase Backend-as-a-Service | PostgreSQL 15 + RLS | Material 3 Dairy Design | Trilingual (EN / HI / MR) | Direct UPI Payouts

PROJECT NAME
MilkDirect (Local Fresh Milk Marketplace)

VERSION / BUILD
Release v1.0.0 (Build 1)

TARGET PLATFORMS
Android (APK), iOS, Desktop & Mobile Web

CORE ARCHITECTURAL ROLES
Customer, Supplier (Farmer/Milkman), Administrator

DATABASE ISOLATION PREFIX
mk_ (Pre-configured for shared Supabase tenants)

LOCALIZATION MATRIX
English, Hindi (हिंदी), Marathi (मराठी)

Confidential & Proprietary — Antigravity Engineering

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1. Executive Summary & Project Vision

MilkDirect is a purpose-built hyperlocal dairy marketplace engineered to disintermediate the conventional dairy distribution chain. In traditional supply chains, milk travels through multiple middlemen, chilling centers, and industrial processing plants, taking 48 to 72 hours to reach urban households. This industrial cycle strips the milk of its natural flavor, promotes adulteration risks, and severely reduces the profit margin of grassroot dairy farmers.

⚠️ The Conventional Dairy Crisis

- **Delayed Delivery:** Packaged pouch milk is often 2–3 days old before reaching doorsteps.
- **Middleman Margins:** Aggregators and brands extract up to 45% of consumer retail value.
- **Opaque Nutrition:** Fat & SNF levels are standardized artificially by blending powdered solids.
- **Severe Adulteration:** Chemical preservation agents and synthetic stabilizers are common.

🥛 The MilkDirect Paradigm

- **Sub-4-Hour Milking-to-Door:** Morning milking delivered by 7:00 AM; Evening by 7:00 PM.
- **100% Farmer Margin:** Direct UPI payouts from consumer to farmer with zero commission cut.
- **Complete Transparency:** Lab-tested Fat% & SNF% declared per cattle listing.
- **Trust & Accountability:** Hyperlocal customer review aggregate and admin FSSAI audit.

> **Key Strategic Differentiator:** Unlike grocery quick-commerce platforms that operate expensive dark stores and high-inventory warehouses, MilkDirect operates as an *asset-light, decentralized marketplace*. The platform manages ordering logic, subscriptions, verification, and route schedules, while physical delivery is executed directly by local farmers and village milkmen.

2. System Architecture & Technology Stack

MilkDirect is built using a modern, reactive, mobile-first client architecture integrated with a cloud-native serverless backend. The application guarantees sub-second responsiveness, seamless offline caching for role selection, and bank-grade data isolation.

1. Presentation Tier
Flutter 3.3+ (Dart)
Material 3 Custom Dairy Theme
Mobile (Android/iOS) + Web

2. State & Gatekeeper
Provider State Management
Role-based AuthGate
SharedPreferences Cache

3. Cloud Services
Supabase BaaS
Supabase Auth (Email/Pass)
Supabase Storage Buckets

4. Persistence & Security
PostgreSQL 15 (`mk_` prefix)
Row Level Security (RLS)
Automated SQL Triggers

Technology Component Matrix

LAYER / DOMAIN	TECHNOLOGY / LIBRARY	ARCHITECTURAL PURPOSE & RESPONSIBILITY
Cross-Platform Core	`Flutter SDK (>=3.3.0 <4.0.0)`	Single codebase compiled natively to Android APK, iOS binary, and Web HTML5/CanvasKit.
State Architecture	`provider (^6.1.2)`	Reactive state management via `AppState` and `ChangeNotifier` listening to session changes.
Cloud Engine	`supabase_flutter (^2.8.0)`	Authentication, PostgreSQL queries, real-time subscription synchronization, and storage access.
Typography & Theme	`google_fonts (^6.2.1)`	Material 3 theme utilizing Poppins (headings) and Nunito (body text) for high readability.
Scheduling & Delivery	`table_calendar (^3.1.2)`	Visual delivery calendar displaying delivered, pending, and missed milk deliveries dynamically.
Offline Persistence	`shared_preferences (^2.3.0)`	Local cache storing the last chosen role (`last_role`) and selected language locale.
Visual Micro-Animations	`shimmer (^3.0.0)`	Skeleton loaders rendering during asynchronous data fetch across tabs and cards.
Image Handling	`cached_network_image (^3.4.1)`	Aggressive multi-tier caching of dairy cattle, farm photos, and supplier avatars.
Communications	`url_launcher (^6.3.1)`	Native invocation of phone calls, SMS, and direct WhatsApp chats with delivery personnel.

3. Multi-Role Identity & Gateway System

Security and role segregation are foundational to MilkDirect. Users interact with the platform through three distinct personas: **Customer**, **Supplier (Farmer / Milkman)**, and **Administrator**.

CUSTOMER

Explores nearby dairy farms, manages delivery addresses, customizes recurring daily milk subscriptions, tracks deliveries, and pays suppliers directly via UPI.

SUPPLIER

Configures cattle listings (Cow/Buffalo), pricing, fat/SNF stats, delivery PIN codes, and executes daily dispatch routes with 1-tap delivery confirmations.

ADMINISTRATOR

Verifies supplier KYC and FSSAI credentials, moderates user accounts, arbitrates dispute tickets, and monitors platform-wide business analytics.

Strict Role-Enforcement & Security Controls

> **Cross-Role Collision Prevention:** A registered Supplier cannot log in through the Customer portal, and a registered Customer cannot log in through the Farmer/Milkman portal. If a user attempts a mismatched role login, the client instantly executes a clean sign-out and notifies the user with role-specific guidance:
> *"This email is registered as a Supplier. Please use the Farmer / Milkman Login."*

- **Role Persistence & Instant Switching:** The app stores the chosen portal in `SharedPreferences` (`last_role`). On subsequent launches, it bypasses the role selection screen and loads the relevant login immediately. Tapping "Change Role" instantly flushes the local preference and returns to the master entry gate.
- **Live 3-Segment Password Strength Meter:** Built into all registration forms, providing real-time visual feedback: **WEAK (<8 CHARS OR NUMERIC ONLY)** → **FAIR (MIXED CHARS)** → **STRONG (8+ CHARS ALPHANUMERIC)**.
- **Zero In-App Admin Registration:** To prevent unauthorized escalation, admin registration forms are completely absent from the client. Administrators are provisioned exclusively by direct database administrator intervention in the Supabase console.
- **Account Suspension (`is_blocked`):** If any user account is flagged for fraud or malicious behavior, administrators can toggle `is_blocked = true`, immediately aborting their current session and barring future logins.

4. Deep-Dive Portal Specifications

4.1 Customer Experience Portal

Designed for frictionless discovery, automated morning delivery, and complete visibility into milk provenance:

🔍 Hyperlocal Supplier Discovery

- **PIN Code Filtering:** Only shows suppliers serving the customer's active delivery PIN code.
- **Cattle & Milk Type Filtering:** 1-tap filters for Pure Cow Milk, Buffalo Milk, or Mixed.
- **Processing Transparency:** Labels Raw (unboiled), Boiled, or Pasteurised milk.
- **Quality Badges:** Displays certified Fat %, SNF %, distance, and average review stars.

🗓️ Recurring Subscription Engine

- **Custom Delivery Schedules:** Daily, Alternate Days, or Custom Weekdays (e.g. Mon-Wed-Fri).
- **Configurable Quantity:** 0.5L, 1.0L, 1.5L, 2.0L+ per delivery slot.
- **Delivery Slot Choice:** Morning (5:00–8:00 AM) or Evening (5:00–8:00 PM).
- **Pause & Resume:** Vacation mode allows 1-tap suspension of deliveries without canceling.

📍 Smart Multi-Address Book

- **Multiple Delivery Points:** Store Home, Office, or Parents' addresses.
- **Precision Coordinates:** Captures latitude/longitude for optimal delivery routing.
- **Detailed Landmarks:** Flat/House number, street, society, and local landmark fields.

💳 Billing & Dispute Ticketing

- **Direct UPI Payments:** Monthly ledger with direct link to supplier's UPI VPA.
- **Delivery Audit Calendar:** Interactive calendar showing delivered vs missed days.
- **Complaint Desk:** Raise dispute tickets for sour milk, late arrival, or quantity issues.

4.2 Supplier (Farmer & Milkman) Portal

Engineered to empower rural cattle owners with digital route management, listing autonomy, and immediate cash flow:

📋 4-Step Registration Wizard

- **Step 1 (Identity):** Classification as *Farmer (Own Cattle)* vs *Milkman (Village Sourced)*, farm name, and contact details.
- **Step 2 (Service Area):** Farm village, home PIN code, and multi-PIN service delivery coverage list.
- **Step 3 (Payouts & Slots):** Direct UPI ID, active delivery slots, minimum order quantity, and delivery fee.
- **Step 4 (Quality & Legal):** Optional FSSAI license upload, hygiene declaration, and purity commitment.

🚚 Daily Dispatch & Execution Sheet

- **Route Organization:** Morning and Evening delivery runs sorted by address sequence.
- **1-Tap Status Logging:** Quick action buttons to mark each bottle as `Delivered` or `Missed`.
- **Direct Customer Dialing:** Integrated call and WhatsApp buttons to resolve delivery issues.
- **Realtime Yield Metrics:** Daily litres required calculation for morning milking scheduling.

4.3 Administrator Oversight Portal

Provides complete administrative governance, verification vetting, and community dispute arbitration:

🛡️ Supplier KYC & FSSAI Verification

Pending supplier profiles are held in an unverified state (hidden from consumer search) until the admin reviews their farm credentials, FSSAI numbers, and declarations. Admin can **Approve** or **Reject with a custom explanation**.

⚖️ Dispute Arbitration & User Safety

Centralized inbox of customer complaints. Administrators inspect order history, contact both parties, issue resolutions, and have the power to instantly suspend or block fraudulent profiles (`is_blocked`).

5. Database Architecture & Data Schema

MilkDirect runs on PostgreSQL inside Supabase. All tables, functions, triggers, and foreign keys utilize the `mk_` namespace prefix. This allows MilkDirect to coexist harmoniously in shared Supabase environments alongside other projects (such as *queuekissan*) without naming collisions.

Complete Data Dictionary (9 Core Tables)

TABLE NAME	PRIMARY KEYS & FOREIGN KEYS	KEY COLUMNS, DATA TYPES & FUNCTIONAL ROLE
`mk_profiles`	`id (uuid, PK → auth.users)`	`name (text), phone (text, unique), alternate_phone (text), role ('customer'|'supplier'|'admin'), language ('en'|'hi'|'mr'), is_blocked (boolean)`. Master user account entity.
`mk_supplier_profiles`	`id (uuid, PK)` `user_id (uuid → mk_profiles)`	`shop_name (text), type ('farmer'|'milkman'), village (text), pincode (text), upi_id (text), fssai_no (text), verification_status ('pending'|'verified'|'rejected'), rating_avg (numeric), delivery_slots (text[])`.
`mk_products`	`id (uuid, PK)` `supplier_id (uuid → mk_supplier_profiles)`	`milk_type ('cow'|'buffalo'|'other'), price_per_litre (numeric), daily_quantity_l (numeric), fat_percent (numeric), snf_percent (numeric), source ('own_cattle'|'collected'), processing ('raw'|'boiled'|'pasteurised'), available (bool)`.
`mk_addresses`	`id (uuid, PK)` `user_id (uuid → mk_profiles)`	`label (text), line (text), landmark (text), pincode (text), contact_phone (text), is_default (bool), latitude (numeric), longitude (numeric)`. Customer delivery locations.
`mk_service_areas`	`id (uuid, PK)` `supplier_id (uuid → mk_supplier_profiles)`	`pincode (text, not null)`. Defines geographic delivery PIN codes served by each registered supplier for hyperlocal filtering.
`mk_subscriptions`	`id (uuid, PK)` `user_id → mk_profiles` `product_id → mk_products`	`address_id → mk_addresses, quantity_l (numeric), schedule_type ('daily'|'alternate'|'weekdays'), weekdays (int[]), slot ('morning_5_8'|'morning_8_10'|'evening_5_8'), status ('active'|'paused'|'cancelled')`.
`mk_orders`	`id (uuid, PK)` `subscription_id → mk_subscriptions` `supplier_id → mk_supplier_profiles`	`order_date (date), quantity_l (numeric), amount (numeric), status ('scheduled'|'delivered'|'missed'|'cancelled'), slot (text), delivered_at (timestamptz)`. Daily delivery fulfillment tracking.
`mk_reviews`	`id (uuid, PK)` `user_id → mk_profiles`	`rating (int, 1-5), comment (text), created_at (timestamptz)`. Triggers automated database recalculation of supplier's `rating_avg` and `rating_count`.

TABLE NAME	PRIMARY KEYS & FOREIGN KEYS	KEY COLUMNS, DATA TYPES & FUNCTIONAL ROLE
	`supplier_id → mk_supplier_profiles`	
`mk_complaints`	`id (uuid, PK)` `user_id → mk_profiles` `order_id → mk_orders`	`category (text), description (text), status ('open'|'investigating'|'resolved'|'dismissed'), resolution (text)`. Customer dispute tickets arbitrated by administrators.

Row Level Security (RLS) & Multi-Tenant Isolation

> Supabase Row Level Security (RLS) is strictly enabled across all `mk_` tables. Customers can only read and write their own addresses, subscriptions, and complaints. Unverified suppliers are hidden from public product queries via `verification_status = 'verified'` filters. Only users with `mk_profiles.role = 'admin'` are granted global update permissions.

6. Material 3 Design System & Aesthetics

The visual architecture of MilkDirect is rooted in a fresh, agricultural-modern aesthetic termed the **Milk & Pasture Material 3 Theme** (defined in `lib/core/theme.dart`). It balances dairy purity with lush pasture vitality.

Pasture Green

`#2E7D32` / `#1B5E20`

Represents lush fodder, nature, and organic authenticity. Used for primary CTAs, app headers, and active tabs.

Dairy Blue

`#1E6FD9`

Represents clean dairy water, trust, and consumer reliability. Used for customer accents and delivery status tags.

Butter Warm Yellow

`#F6C453` / `#E5A823`

Represents rich farm butter and ghee. Used for warning badges, star ratings, and subtle surface card highlights.

- **Surfaces & Cards:** Milk White scaffold (`#FDFCF8`) paired with warm Cream surfaces (`#FFF7E6`) and pure white cards with 16–20px corner radii and delicate elevation shadows.
- **Ergonomic Touch Targets:** 54px tall tactile pill buttons (`AppButton`) optimized for one-handed operation by rural delivery personnel.
- **Typography:** Google Fonts combination of **Poppins** for crisp headers and **Nunito** for readable, friendly body copy.

7. Multilingual Localization Engine

To ensure complete accessibility across diverse linguistic demographics—from urban homemakers to rural dairy farmers—MilkDirect implements native, instant, trilingual localization without requiring app restarts (implemented in `lib/core/l10n.dart`).

LANGUAGE	CODE	TARGET DEMOGRAPHIC	REPRESENTATIVE UI TERMS
English	`en`	Urban customers & Platform Administrators	*"MilkDirect", "Suppliers", "My Milk", "Billing", "Verify"*
हिंदी (Hindi)	`hi`	North & Central Indian consumers & milkmen	*"मिल्कडायरेक्ट", "डेयरी वाले", "मेरा दूध", "बिलिंग", "सत्यापित"*
मराठी (Marathi)	`mr`	Maharashtra dairy heartlands & local farmers	*"मिल्कडायरेक्ट", "दूध उत्पादक", "माझे दूध", "बिल", "पडताळणी"*

8. Deployment, Setup & Administration Guide

Step 1: Supabase Configuration

1. Enable **Email Provider** in Supabase Dashboard → Authentication → Providers → Email.
2. Configure password policy to enforce a minimum of 8 characters.
3. Execute `supabase/schema.sql` in the Supabase SQL editor to create all `mk_` tables and triggers.

Step 2: Provisioning the First Master Admin

Admin accounts cannot be created from the public mobile application. They must be provisioned via SQL:

```
-- 1. Create the user in Supabase Auth Dashboard (e.g. admin@milkdirect.com) and copy User UID
-- 2. Execute SQL to provision admin role:
INSERT INTO mk_profiles (id, name, phone, role, language, is_blocked)
VALUES (
  'PASTE_AUTH_USER_UID_HERE',
  'Master Administrator',
  'admin@milkdirect.com',
  'admin',
  'en',
  false
)
ON CONFLICT (id) DO UPDATE SET role = 'admin';
```

Step 3: Compilation & Execution Commands

1. Fetch Packages

`flutter pub get`

Resolves all dependencies including Supabase, Provider, and Material components.

2. Run Web Development

`flutter run -d chrome`

Launches real-time live reload on Google Chrome for rapid UI testing.

3. Build Android Release APK

`flutter build apk --release`

Compiles an optimized, standalone release APK ready for distribution.

9. Quality Assurance & Verification Matrix

MilkDirect adheres to a comprehensive 17-point verification protocol before production release:

#	TEST SCENARIO	ACTION / EXECUTION STEPS	EXPECTED ARCHITECTURAL RESULT
1	**Role Selection Launch**	Launch app with empty SharedPreferences	Displays 2 large cards (Customer, Farmer) + 1 bottom link (Admin).
2	**Role Persistence**	Select Customer → kill & reopen app	Bypasses selector, lands directly on Customer Login with "Change role".
3	**Role Reset Flow**	Click "Change role" on login or profile	Flushes `last_role` cache and returns to Role Selection screen.
4	**Customer Sign-Up**	Register with name, email, phone, PIN & address	Creates Auth user, `mk_profiles` and `mk_addresses`, routes to Home.
5	**Password Strength Meter**	Type passwords of varying complexity	Segments react in real-time: Weak (red), Fair (amber), Strong (green).
6	**Supplier Onboarding**	Complete 4-step wizard (Farm, PINs, UPI, FSSAI)	Saves to `mk_supplier_profiles`; holds in `VerificationPendingScreen`.
7	**Cross-Role Block (Supplier → Cust)**	Supplier email entered on Customer Login	Instantly logs out with: "*Registered as Supplier. Use Farmer Login.*"
8	**Cross-Role Block (Cust → Supplier)**	Customer email entered on Supplier Login	Instantly logs out with: "*Registered as Customer. Use Customer Login.*"
9	**Admin Access Security**	Non-admin email entered on Admin Login	Signs out immediately: "*Access Denied: Not an administrator.*"
10	**Supplier Vetting Flow**	Admin taps "Approve" on pending supplier	Status transitions to `verified`; listings go live to nearby consumers.
11	**Dispute Ticket Resolution**	Customer files complaint; Admin resolves	Ticket status updates to `resolved`; logged in customer billing tab.
12	**Language Switching**	Toggle EN → HI → MR in app bar	Instant UI string re-render across forms, buttons, and error banners.

10. Future Strategic Roadmap

🔬 IoT Milk Analyzer Sync

Integration with Bluetooth digital ultrasonic lactometers at dairy collection booths to automatically sync Fat% and SNF% directly to Supabase without manual entry.

🗺️ GPS Route Optimization

Automated traveling salesperson algorithm generating turn-by-turn navigation for morning milk delivery riders, reducing delivery duration by up to 35%.

🔄 UPI Autopay / e-NACH

Automated recurring monthly bank debits via National Payments Corporation of India (NPCI) UPI Autopay, eliminating manual monthly billing follow-ups.